

 AI ENGINEERING DECISIONS

Finetuning

VS

RAG

VS

Agentic AI

Know which to use.
Know when to combine them.



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! THE PROBLEM

Most Teams Pick the Wrong Tool

These three are not interchangeable. Using the wrong one wastes time, money, and accuracy.



Finetuning when RAG would solve it faster and cheaper



RAG when the model needs a behavior change, not more knowledge



Agents when a simple pipeline would be more reliable



Each wrong choice multiplies cost and complexity

→ The right tool depends on the problem, not the trend.



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Finetuning

What it is: Training the base model on your labeled data to permanently change how it behaves.

HOW FINETUNING WORKS

Raw
Data



Label
Examples



Train
Model



Tuned
Model



Deploy &
Serve

✓ BEST FOR

- Specific tone or writing style
- Domain-specific jargon
- Consistent output format
- High-volume repeated tasks

⚠ LIMITATIONS

- ✗ Needs lots of labeled data
- ✗ Expensive to retrain
- ✗ Knowledge goes stale
- ✗ Slow iteration cycles



Use it when behavior needs to be baked in, not retrieved.



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RAG

What it is: Fetching relevant documents at runtime and injecting them into the model's context before it answers.

🔧 HOW RAG WORKS

📄 INGESTION PIPELINE



⚡ QUERY PIPELINE



✅ BEST FOR

- Real-time or changing knowledge
- Large document corpora
- Auditable, cited answers
- Private enterprise data

★ KEY ADVANTAGES

- ✓ No retraining needed
- ✓ Always up to date
- ✓ Sources are traceable
- ✓ Easy to update



Use it when knowledge needs to stay current and explainable.



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Agentic AI

What it is: AI that plans, decides, and executes multi-step tasks using tools, memory, and self-correction.

HOW AGENTS WORK



✓ BEST FOR

- Multi-step workflows
- Open-ended tasks
- Tasks requiring real actions
- Autonomous execution

⚠ WATCH OUT FOR

- ✗ Hard to debug
- ✗ High cost per run
- ✗ Needs strong guardrails
- ✗ Slowest to complete



Use it when the task cannot be solved in a single inference step.



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Side by Side

FINETUNING	RAG	AGENTIC
COST High upfront	COST Low per query	COST High per run
FRESHNESS Static	FRESHNESS Real-time	FRESHNESS Real-time
SPEED Fast inference	SPEED Adds latency	SPEED Slowest
DATA NEEDED Labeled pairs	DATA NEEDED Documents	DATA NEEDED Tool definitions
COMPLEXITY Medium	COMPLEXITY Medium-High	COMPLEXITY Very High
BEST WHEN Behavior change	BEST WHEN Knowledge needed	BEST WHEN Multi-step tasks



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How to Choose

Finetune

→ when you need a specific **tone, format, or behavior** applied consistently

RAG

→ when the model needs access to **specific, current knowledge**

Agentic

→ when the task requires **multiple steps and real-world actions**



ASK FIRST

**Behavior or format
problem?**

FINETUNE



THEN ASK

**Knowledge or
accuracy problem?**

RAG



FINALLY ASK

**Execution or workflow
problem?**

AGENTIC



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⚠ WATCH OUT

Common Mistakes

These are the traps most teams fall into when choosing their AI strategy.



Finetuning when data changes frequently, making retraining a constant cost



Using RAG to fix a tone or format issue (retrieval cannot change behavior)



Building agents for simple, deterministic tasks that add risk and cost



Treating these as mutually exclusive when they are designed to be combined



Define the problem clearly first. Then pick the tool.



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How They Stack Together

These three are not competing. They are complementary layers of a production AI system.

ARCHITECTURE: STACKED LAYERS

**Agentic Layer**What it can **DO****RAG Layer**What it **KNOWS****Finetuning Layer**How it **BEHAVES**

FINETUNING + RAG

A domain-tuned model that retrieves fresh knowledge. It speaks your language and stays current without retraining.

RAG + AGENTIC

Agents that retrieve relevant context before acting. More grounded decisions, fewer hallucinated steps.

ALL THREE TOGETHER

Specialized behavior, grounded in live knowledge, executing complex workflows. This is production-grade AI.

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💡 SUMMARY

The Takeaway

Finetune → changes how the model **behaves**

RAG → changes what the model **knows**

Agentic → changes what the model can **do**



FINETUNE ONLY
Consistent Behavior



ADD RAG
Grounded Knowledge



ADD AGENTS
Autonomous Action

Production AI = Right Tool + Right Problem + Evaluation

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